Linux Operating Systems

Assignment

Operating System, Linux Distributions and Exit Status

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1. What If There Were No Operating System? Imagine a computer without an operating system. Identify the problems a user would face and investigate how OS features solve them. Demonstrate at least three features using Linux and present a *"Without OS vs With OS" comparison.

A computer without an operating system is just hardware. It cannot run programs, save files, or use the keyboard and screen. The operating system is what makes all of this possible.

Some problems a user would face without an OS:

- Only one program can run at a time.
- No file system, so there is no way to save or organize data.
- No security, since any program can access any data.
- No memory management, so programs would crash often evenly.

Three OS features shown using Linux

Problem Without OS	How OS Solves It	Linux Command Used
Cannot run many programs together	OS shares CPU time between programs	ps - shows running programs
No place to store or organize files	OS gives a file system with folders	mkdir, ls - create and list files
No protection for memory or data	OS gives each program its own memory space and sets file permissions	free, chmod - check memory, set permissions

Table 1: OS features and the Linux commands that show them

Without OS vs With OS

Point	Without OS	With OS
Running programs	Only one program can run	Many programs run together
Files	No files or folders exist	Files are organized in folders
Memory	Programs overwrite each other	Each program gets its own memory
Security	No protection at all	Permissions protect files
Ease of use	Very hard, needs deep knowledge	Simple commands or clicks

Table 2: Without OS vs With OS

2. Imagine your team is setting up a cybersecurity laboratory. Explore at least five Linux distributions relevant to cybersecurity and recommend suitable distributions for different roles such as penetration testing, digital forensics, privacy, and security monitoring. Present your findings creatively as a comparison chart, poster, infographic, or "OS selection guide."

A cybersecurity lab needs different tools for different jobs, like testing, forensics, privacy work, and monitoring. Below are five Linux distributions suited for these roles.

Distribution	Used For	Why
Kali Linux	Penetration Testing	Comes with ready-made hacking/testing tools like Nmap and Metasploit
Parrot OS	Testing + Privacy	Has testing tools plus Tor and encryption for privacy
CAINE	Digital Forensics	Protects evidence by not writing to the disk being checked
Tails	Privacy	Forces all traffic through Tor and forgets everything after shutdown
Security Onion	Monitoring	Watches network traffic and shows the alerts for threats

Table 3: Linux distributions for cybersecurity roles

3. Exit status: Execute at least five Linux commands that produce both successful and unsuccessful results. Investigate their exit status using $?. Create a simple visual/table showing *Command → Result Exit Status → Meaning*. What can a script learn from an exit status?

After every Linux command runs, it gives a small number called the exit status. This can be checked with the command echo $?. A status of 0 means success, and any other number means failure.

Command	Exit Status	Meaning
ls /home	0	Success - folder found
ls /wrongfolder	2	Failure - folder not found
mkdir demo	0	Success - folder created
rm missing.txt	1	Failure - file not found
grep hello file.txt	0	Success - word found

Command	Exit Status	Meaning
grep xyz file.txt	1	Failure - word not found

Table 4: Commands and their exit status

What a script learns from exit status

A script cannot see the screen, so it uses the exit status to know if the last command worked. Using this, a script can:

- Stop or continue based on success or failure.
- Show an error message or retry if a command fails.
- Run one command only if another one succeeded, using && or ||.